

KRISHNAMAL RAMASUBBAIYER SCHOOL

T.V.R NAGAR, MADURAI

The cover is a vibrant collage of science-related illustrations. At the top, a blue banner contains the school's name. Below it, various scientific symbols are scattered: a blue microscope, a flask with green liquid, a Bohr-style atomic model, a glowing yellow lightbulb, a molecular model, a stack of books labeled 'PHYSICS', 'CHEMISTRY', and 'BIOLOGY', a flask with pink liquid, a globe, and a potted plant. The central text 'SCIENCE PORTFOLIO' is rendered in large, bubbly, multi-colored letters. The bottom features an open book with an atomic diagram, test tubes with colored liquids, and a flask with blue liquid. The entire background is a soft, colorful gradient with small star and flower motifs.

SCIENCE PORTFOLIO

Name : S.Vairalingam

Grade : X-B

Academic Year : 2026-2027

Roll No. : _____

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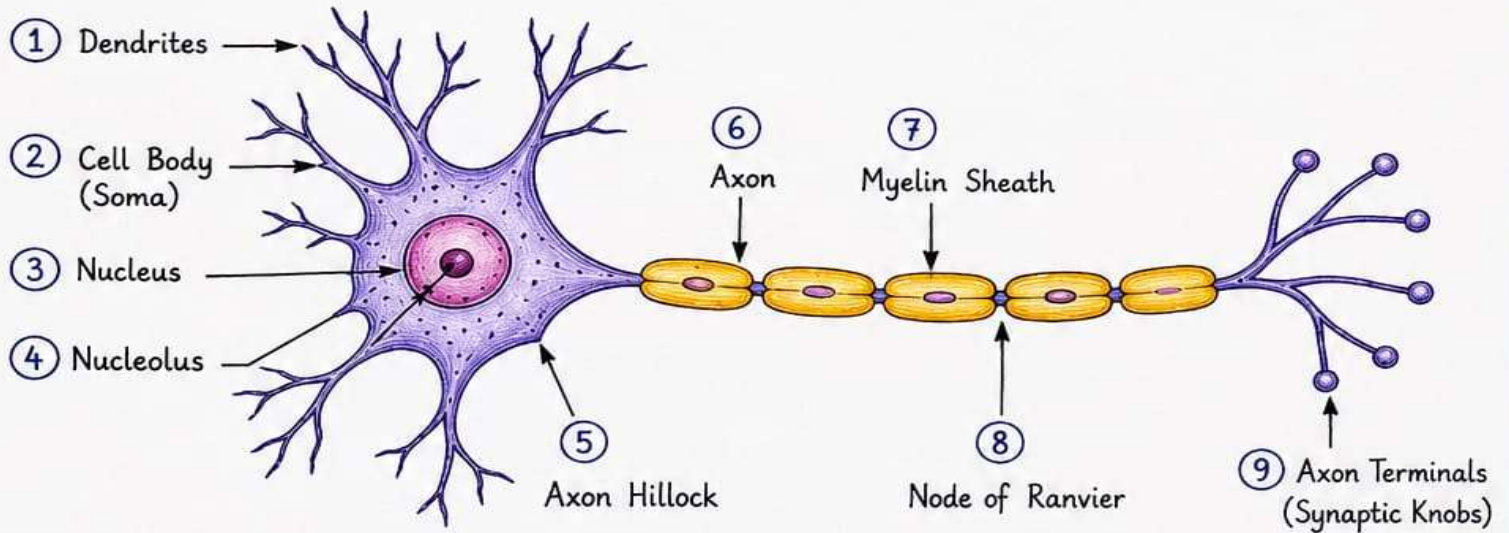
BIOLOGY

— The study of life —

“Exploring life,
from the smallest cell to
the greatest ecosystem.”

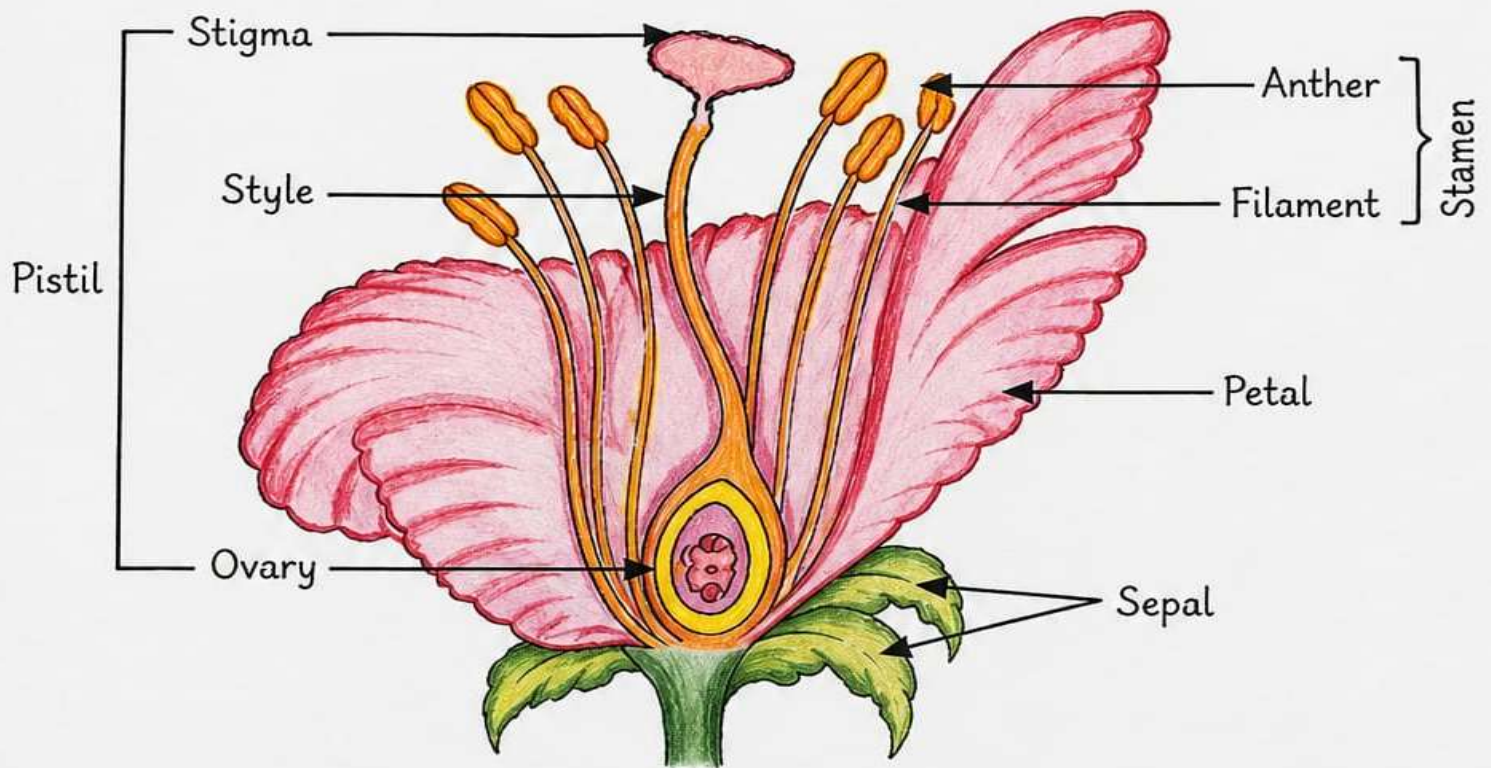


STRUCTURE OF NEURON



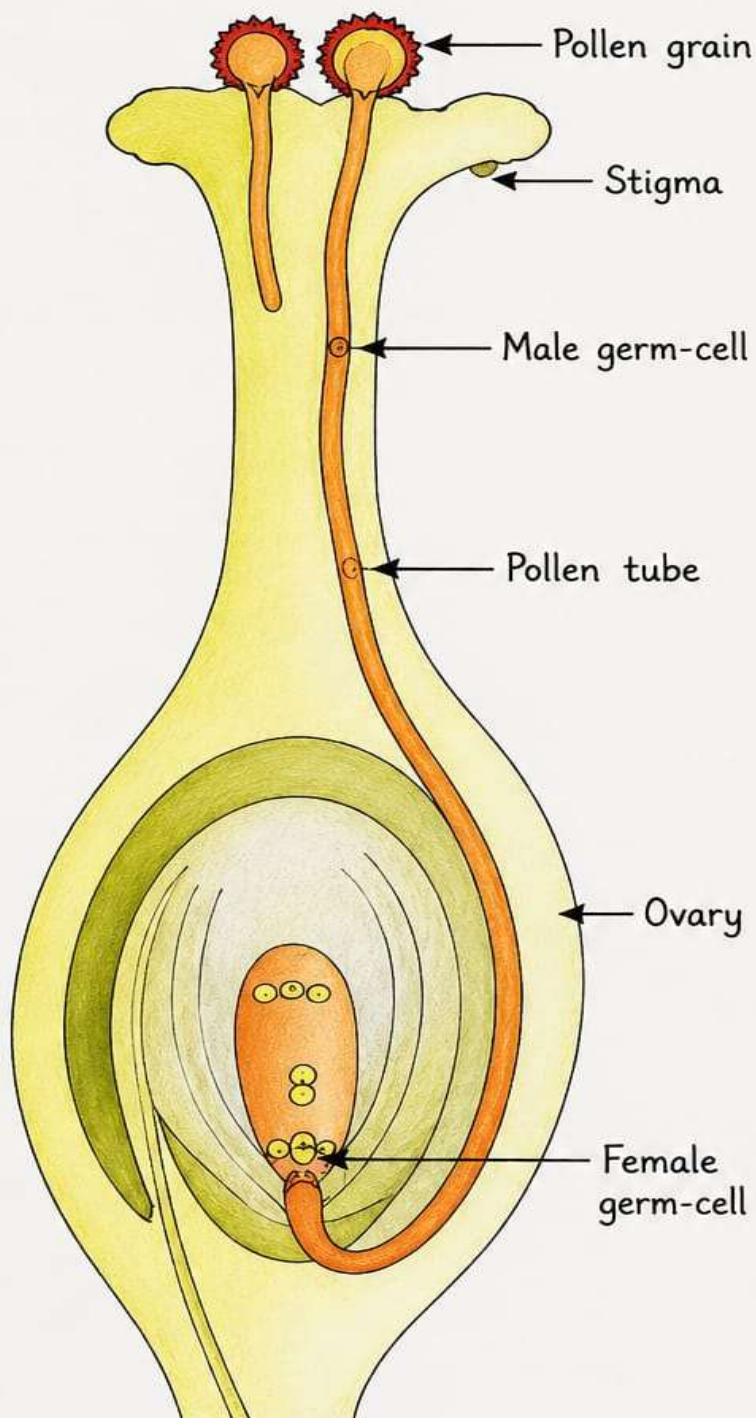
PART	FUNCTION
1. Dendrites	Receive signals or impulses from other neurons or receptors and carry them towards the cell body.
2. Cell Body (Soma)	Integrates the received signals and maintains the cell's metabolic activities.
3. Nucleus	Controls all the activities of the neuron.
4. Nucleolus	Involved in the production of ribosomes.
5. Axon Hillock	Joins the cell body to the axon and helps in the initiation of nerve impulses.
6. Axon	Carries nerve impulses away from the cell body to other neurons or effectors.
7. Myelin Sheath	Insulates and protects the axon and increases the speed of impulse transmission.
8. Node of Ranvier	Gaps between myelin sheaths; important for faster conduction of impulses.
9. Axon Terminals (Synaptic Knobs)	Transmit the nerve impulse to the next neuron, muscle or gland through synapses.

LONGITUDINAL SECTION OF A FLOWER



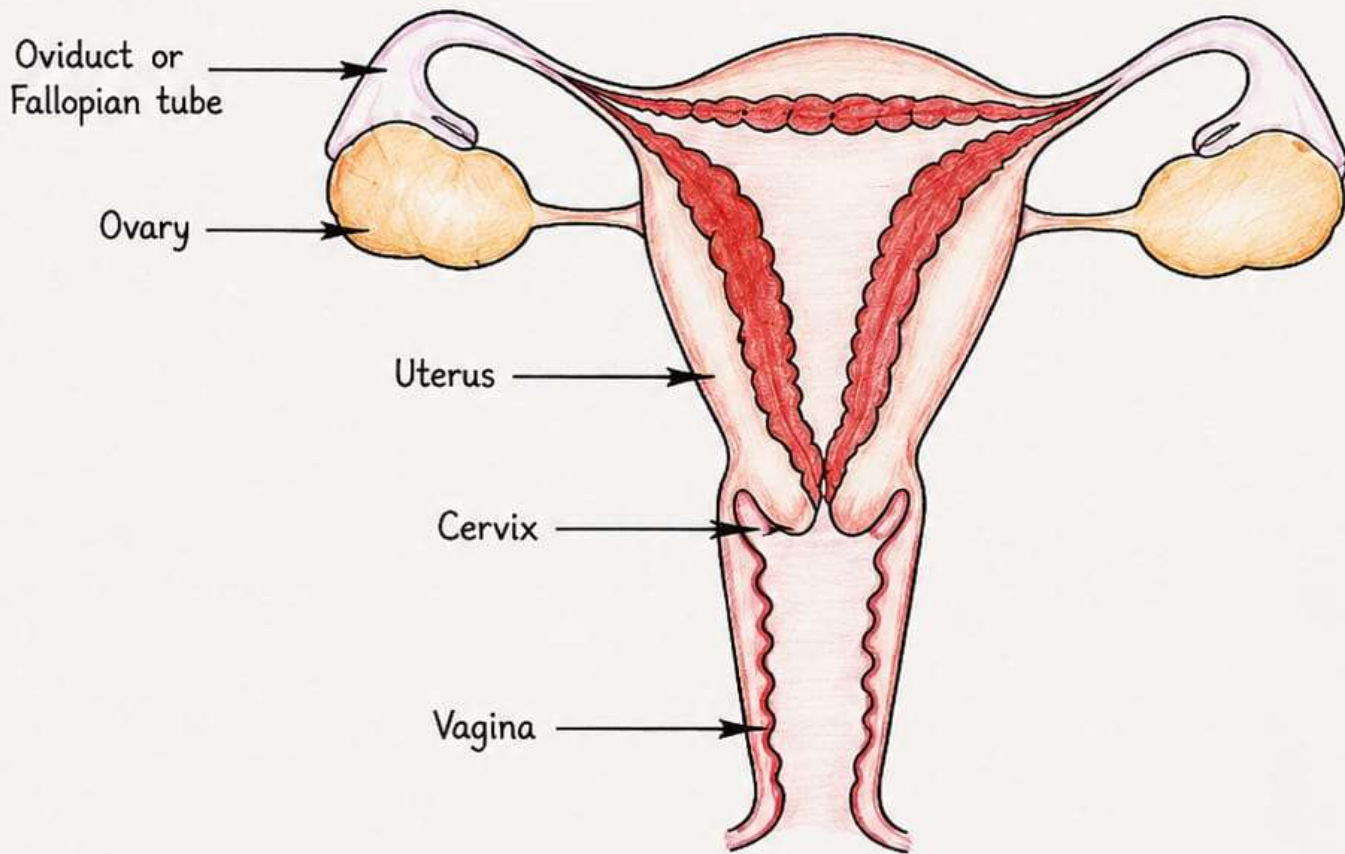
PART	FUNCTION
1. Stigma	Receives pollen grains during pollination.
2. Style	Provides a passage for the pollen tube to reach the ovary.
3. Anther	Produces pollen grains.
4. Filament	Supports the anther and holds it in position.
5. Petal	Attracts pollinators with its colour, scent and nectar.
6. Ovary	Contains ovules; after fertilization, the ovary develops into fruit and ovules into seeds.
7. Sepal	Protects the flower in the bud stage.

GERMINATION OF POLLEN ON STIGMA



PART	FUNCTION
Pollen grain	The male gametophyte produced in the anther which is transferred to the stigma.
Stigma	The receptive surface of the pistil where pollen grains land during pollination.
Male germ-cell	The generative nucleus inside the pollen tube which divides to form two male gametes.
Pollen tube	A tube formed from the pollen grain that grows through the style towards the ovary carrying the male germ-cells.
Ovary	The basal part of pistil containing ovules where fertilization takes place.
Female germ-cell	The egg cell present inside the ovule which fuses with a male gamete during fertilization.

HUMAN FEMALE REPRODUCTIVE SYSTEM



PART	FUNCTION
1. Oviduct or Fallopian tube	• Transports the ovum from the ovary to the uterus.
2. Ovary	• Produces ova (female gametes) and secretes the hormones oestrogen and progesterone.
3. Uterus	• Site of implantation and development of the embryo into a foetus.
4. Cervix	• Narrow lower part of the uterus which opens into the vagina; it remains closed during pregnancy.
5. Vagina	• Muscular passage that receives the penis during copulation; also serves as the birth canal during childbirth and passage for menstrual flow.

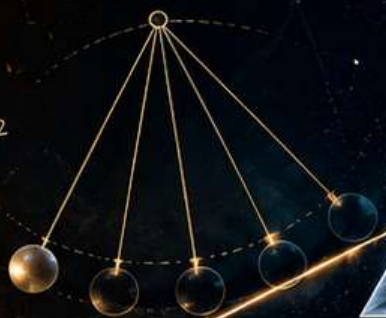
$$F = ma$$

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$E = mc^2$$

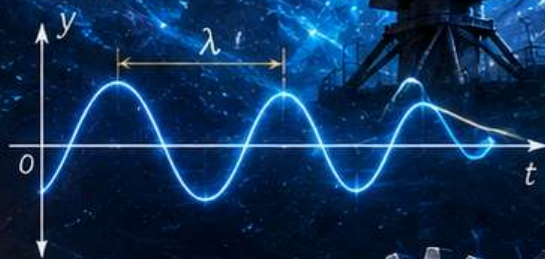
$$p = mv$$



PHYSICS

THE STUDY OF MATTER, ENERGY
AND THE LAWS OF NATURE

UNDERSTAND THE FUNDAMENTALS.
EXPLAIN THE UNIVERSE.
SHAPE THE FUTURE.



MECHANICS



WAVES



ELECTROMAGNETISM



THERMODYNAMICS

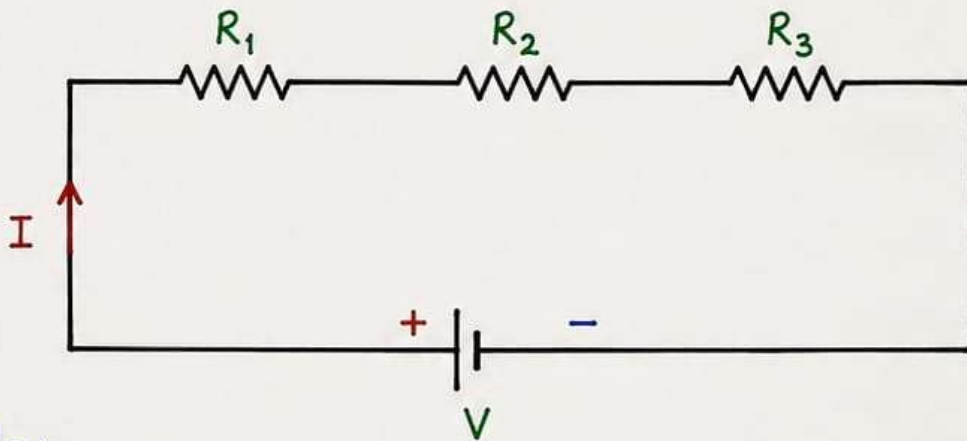


MODERN PHYSICS

RESISTORS IN SERIES

Definition:

When two or more resistors are connected end-to-end so that the same electric current flows through each resistor, they are said to be connected in series.



Formula:

$$R = R_1 + R_2 + R_3 + \dots$$

Where:

R = Total (equivalent) resistance (Ω)

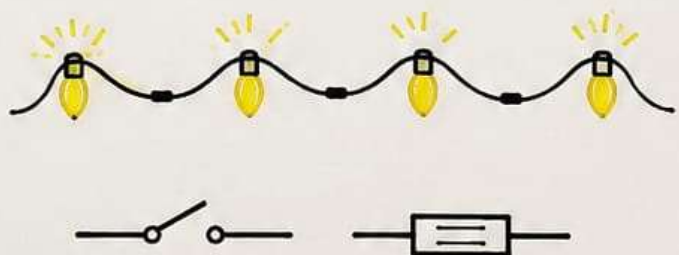
R_1, R_2, R_3 = Individual resistances (Ω)

Characteristics:

- The same current flows through all resistors.
- The total resistance is the sum of all individual resistances.
- The voltage is divided across the resistors.

Applications:

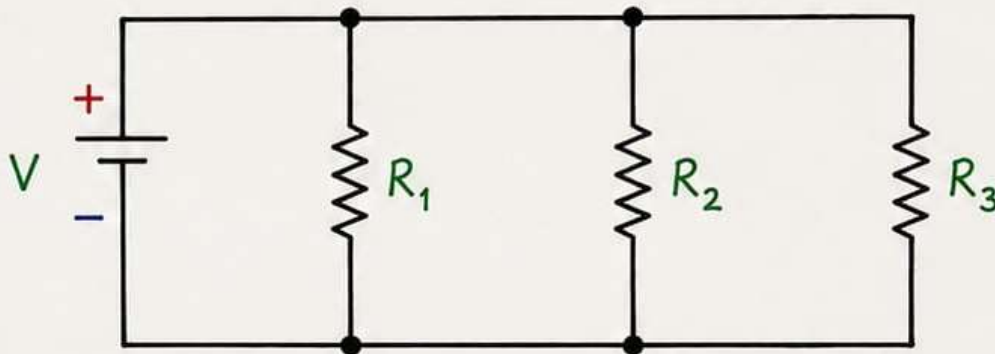
- Decorative light strings
- Voltage divider circuits
- Fuse and switch circuits
- Simple electronic circuits



RESISTORS IN PARALLEL

Definition:

When two or more resistors are connected across the same two points so that the voltage across each resistor is the same, they are said to be connected in parallel.



Formula:

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$

Where:

R = Total (equivalent) resistance (Ω)

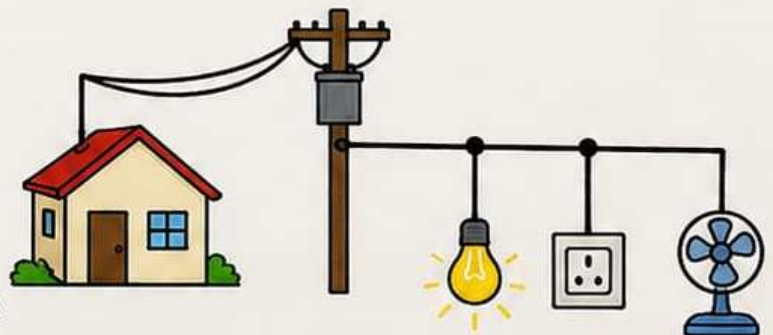
R_1, R_2, R_3 = Individual resistances (Ω)

Characteristics:

- The same voltage is present across all resistors.
- The current divides among the different branches.
- The equivalent resistance is always less than the smallest individual resistance.

Applications:

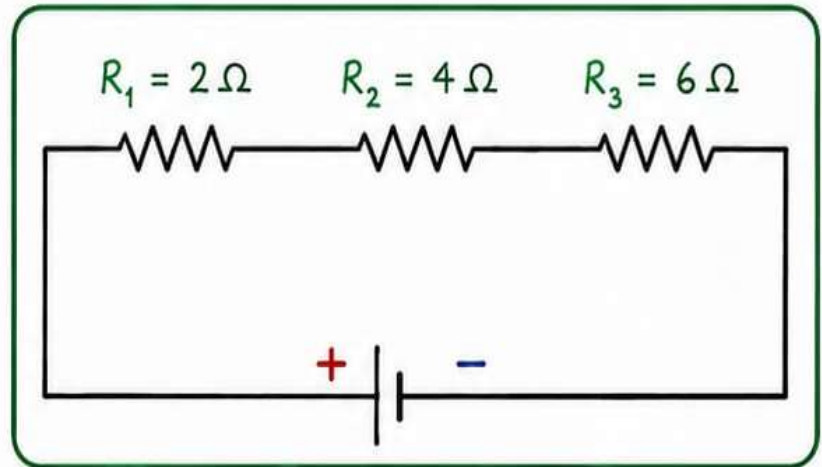
- Household electrical wiring
- Power distribution systems
- Lighting circuits
- Electronic appliances and devices



EXAMPLE – RESISTORS IN SERIES

Question:

Three resistors of $2\ \Omega$, $4\ \Omega$ and $6\ \Omega$ are connected in series. Find the equivalent resistance.



Solution:

Given:

$$R_1 = 2\ \Omega, \quad R_2 = 4\ \Omega, \quad R_3 = 6\ \Omega$$

In series combination,

$$R_{\text{total}} = R_1 + R_2 + R_3$$

$$R_{\text{total}} = 2\ \Omega + 4\ \Omega + 6\ \Omega$$

$$R_{\text{total}} = 12\ \Omega$$

Answer: The equivalent resistance of the series combination is $12\ \Omega$.

Formula:

$$R_{\text{total}} = R_1 + R_2 + R_3$$

(In series)

Where:

R_{total} = Total (equivalent) resistance (Ω)

R_1, R_2, R_3 = Individual resistances (Ω)

Characteristics:

- The same current flows through all resistors.
- The total resistance is the sum of all individual resistances.
- The voltage is divided across the resistors.

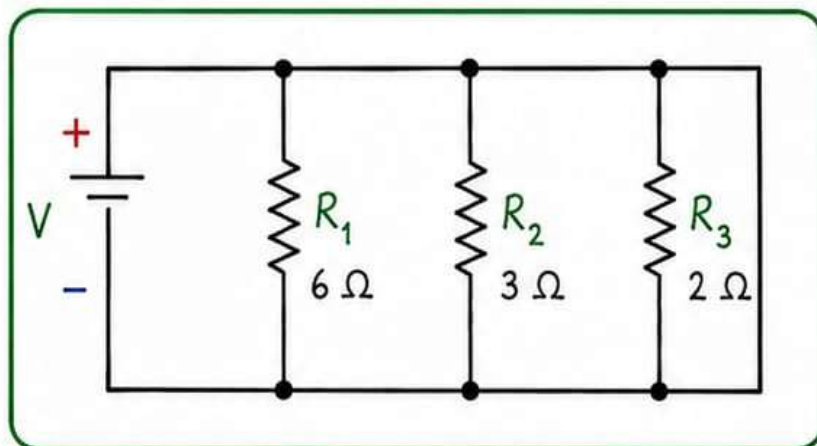
Applications:

- ★ Decorative light strings
- ★ Voltage divider circuits
- ★ Fuse and switch circuits
- ★ Simple electronic circuits

EXAMPLE - RESISTORS IN PARALLEL

Question:

Three resistors of $6\ \Omega$, $3\ \Omega$ and $2\ \Omega$ are connected in parallel across a battery. Find the equivalent resistance.



Solution:

Given:

$$R_1 = 6\ \Omega, \quad R_2 = 3\ \Omega, \quad R_3 = 2\ \Omega$$

In parallel combination,

$$\frac{1}{R_{\text{total}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R_{\text{total}}} = \frac{1}{6} + \frac{1}{3} + \frac{1}{2}$$

$$\frac{1}{R_{\text{total}}} = \frac{1}{6} + \frac{2}{6} + \frac{3}{6}$$

$$\frac{1}{R_{\text{total}}} = \frac{6}{6} = 1$$

$$R_{\text{total}} = 1\ \Omega$$

Formula:

$$\frac{1}{R_{\text{total}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Where:

R_{total} = Total (equivalent) resistance (Ω)

R_1, R_2, R_3 = Individual resistances (Ω)

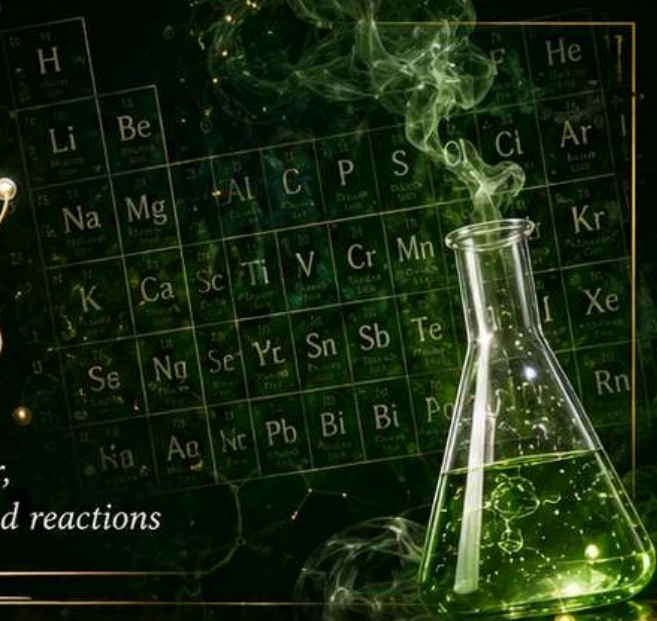
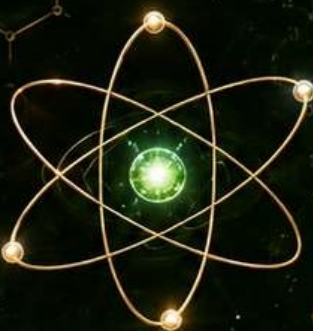
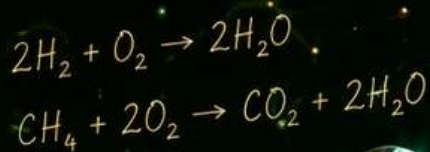
Answer: The equivalent resistance of the parallel combination is $1\ \Omega$.

Characteristics:

- The same voltage is present across all resistors.
- The current divides among the different branches.
- The equivalent resistance is always less than the smallest individual resistance.

Applications:

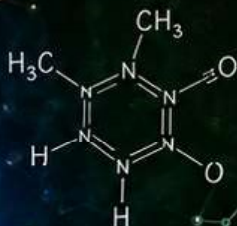
- ★ Household electrical wiring
- ★ Power distribution systems
- ★ Lighting circuits
- ★ Electronic appliances and devices



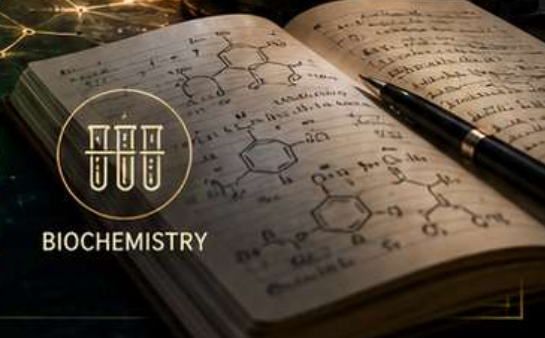
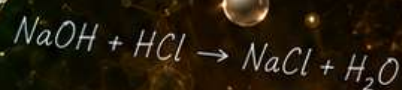
*The study of matter,
its composition, properties and reactions*

CHEMISTRY

EXPLORING THE BUILDING BLOCKS
OF LIFE AND THE UNIVERSE



“ The best way to
predict the future is
to create it. ”



ORGANIC
CHEMISTRY



INORGANIC
CHEMISTRY



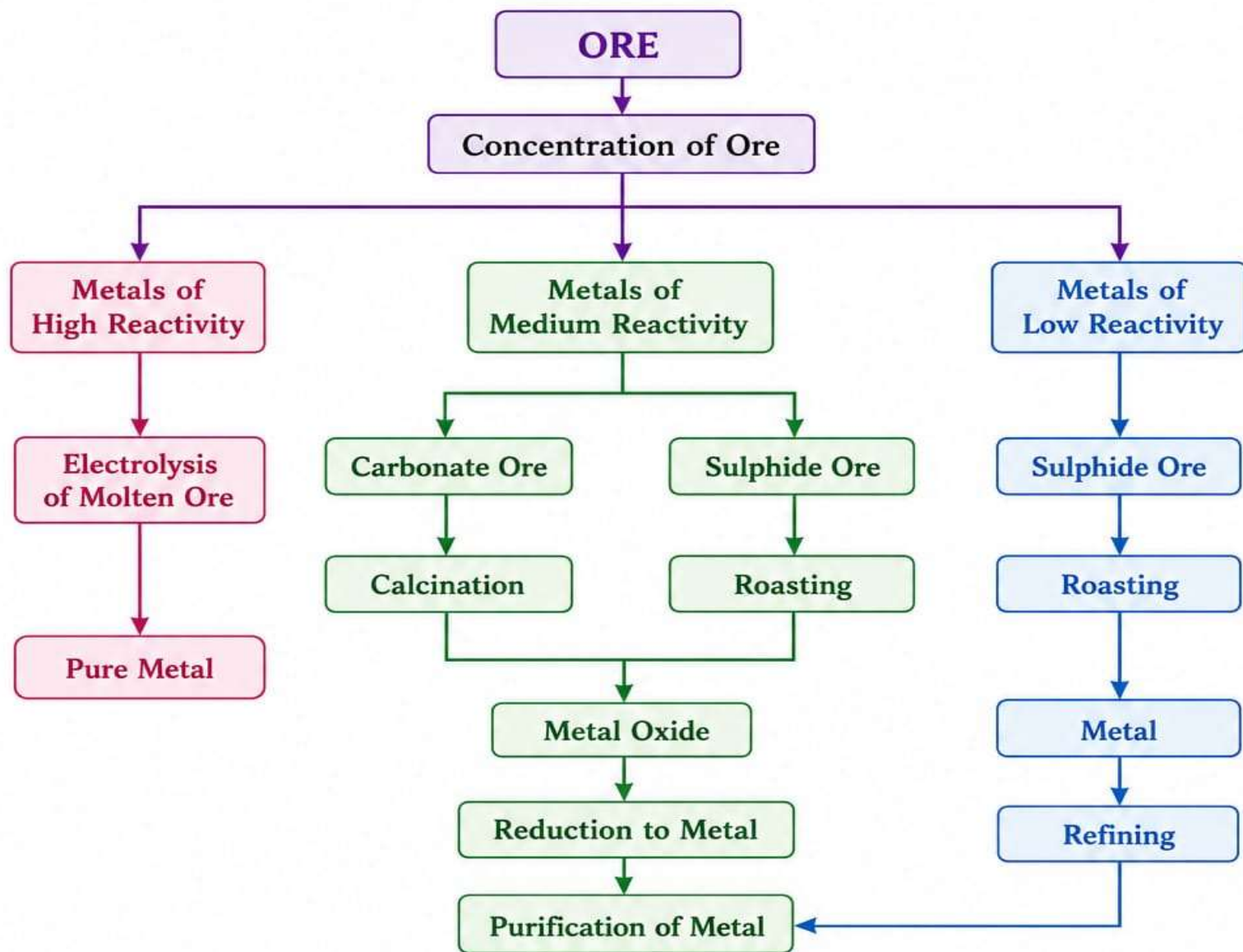
PHYSICAL
CHEMISTRY



ANALYTICAL
CHEMISTRY



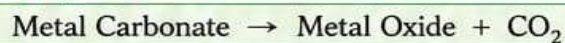
BIOCHEMISTRY



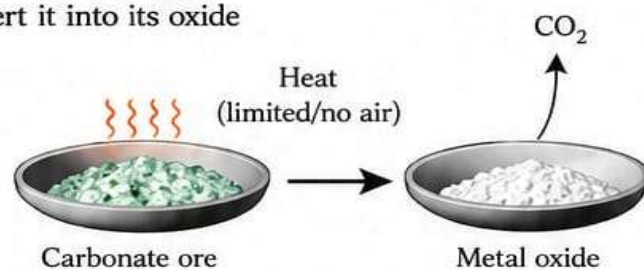
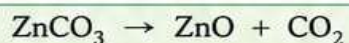
CALCINATION

Definition: Calcination is the process of heating a carbonate ore strongly in the absence or limited supply of air to convert it into its oxide by removing carbon dioxide.

General Equation:



Example:



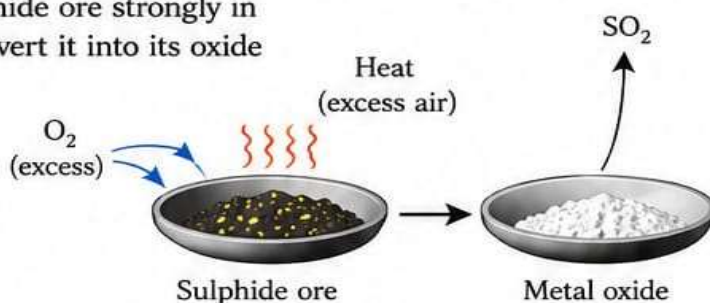
ROASTING

Definition: Roasting is the process of heating a sulphide ore strongly in the presence of excess air (oxygen) to convert it into its oxide by removing sulphur as sulphur dioxide.

General Equation:



Example:



VERSATILE NATURE OF CARBON

Definition

Versatile Nature of Carbon:

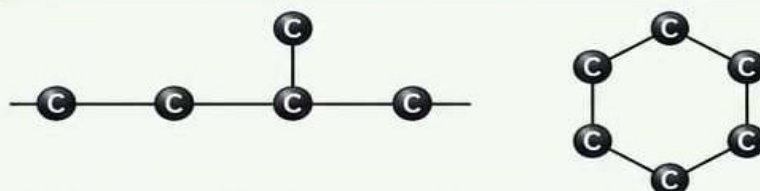
Carbon is called a versatile element because it forms a very large number of compounds due to its **catenation** property and **tetravalency**.

REASONS FOR THE VERSATILE NATURE OF CARBON

1 Catenation

Definition:

Catenation is the property of carbon to bond with other carbon atoms to form long chains, branched chains, and ring structures.

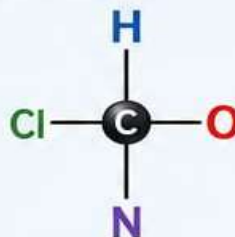


- Forms single, double, and triple bonds.
- Carbon-carbon bonds are strong and stable.
- Results in millions of carbon compounds.

2 Tetravalency

Definition:

Carbon has a valency of 4, so it can form four covalent bonds with carbon or other elements such as hydrogen, oxygen, nitrogen, sulphur, and chlorine.



- Forms stable compounds.
- Produces a wide variety of substances with different properties.

VERSATILE NATURE OF CARBON

